

Concept Note: Cura - An AI-Powered Personal Health Companion

Introduction & Relevance to SDG 3

In alignment with **Sustainable Development Goal 3 (Good Health and Well-being)**, the Cura project aims to leverage artificial intelligence agents to address two critical challenges in personal health management: medication non-adherence and the lack of accessible, personalized nutritional guidance. Many individuals, particularly busy professionals and the elderly, struggle to manage complex medication schedules, while others find it difficult to create diet plans that align with their specific health goals (e.g., weight loss, diabetes management). Cura is an AI-powered platform designed to bridge this gap by providing a proactive, intelligent, and automated solution for daily health management.

Objective

The primary objective of this project is to develop and implement a unified digital health platform that integrates two specialized AI agents—a **Reminder Agent** and a **Diet Agent**—to provide users with seamless, personalized support for medication adherence and nutritional planning.

Rationale

Medication non-adherence is a significant global health issue that leads to poor health outcomes and increased healthcare costs. Simultaneously, generic diet advice often fails to meet the unique needs of individuals with specific health conditions or goals. By leveraging AI, Cura can offer a hyper-personalized solution that automates these critical daily tasks. The platform moves beyond simple static reminders to create a comprehensive health management ecosystem, empowering users to take control of their well-being with intelligent, data-driven assistance.

Methodology & Project Architecture

The Cura platform is built on a **microservices architecture** to ensure scalability and maintainability. The system consists of a central Django backend that orchestrates two independent AI agent services.

1. Core Backend (Django REST Framework): This is the central hub of the application.

It is responsible for:

- Secure user authentication and management.
- Providing a robust REST API for all frontend interactions.
- Storing and managing all user data (profiles, medications, diet plans) in a PostgreSQL database.
- Orchestrating the agent services by triggering them and processing their outputs.

2. Reminder Agent (Flask & SQLAlchemy): This is a dedicated microservice responsible for all medication-related tasks. Its workflow is as follows:

- It is triggered by an automated cron job managed by the Django backend.
- Upon activation, it queries its database to identify all reminders that are currently due.
- It sends formatted email notifications to users.
- It automatically updates the medicine's inventory and sends a separate refill alert if the stock is low.

3. Diet Agent (Langchain & Google Gemini): This agent is responsible for generating personalized meal plans.

- It is triggered when a user requests a new plan via the Django API.
- The Django backend sends the user's complete health profile (age, weight, goals, allergies, etc.) to the agent.
- The agent uses the **Langchain** framework and **Google's Gemini LLM**, constrained by a Pydantic schema, to generate a complete, structured, one-day diet plan in JSON format.

- This JSON object is returned to the Django backend and saved to the user's profile.

4. **Frontend (Streamlit & PWA):** The user interacts with the system through:

- A web application built with **Streamlit** (Python).
- A **Progressive Web App (PWA)** for a native-like experience on Android devices.

Dataset

The "dataset" for this project is the dynamic, user-generated data stored in the application's PostgreSQL database. This includes:

- **User Profiles:** Health metrics, dietary preferences, allergies, and goals.
- **Medication Data:** Medicine names, dosages, inventory counts, and reminder schedules.
- **Adherence Data (Future Scope):** Logs of when users mark their medication as taken.

Implementation & Testing

- **Implementation:** The project is implemented with a full-featured Django REST API, two independent agent services, and a deployment pipeline configured for Render.
- **Testing:** The system is tested end-to-end by:
 1. Using curl commands and a Python test script to hit all API endpoints.
 2. Manually triggering the agent endpoints.
 3. Validating the outcomes by checking for the successful delivery of reminder emails and verifying that data (e.g., medicine inventory) is correctly updated in the database.

Deployment

The entire project is deployed on **Render** using a `build.sh` script and a Procfile. This configuration handles:

- Cloning the main repository and its Git submodules (the agents).
- Installing all dependencies using Poetry.
- Running database migrations.
- Starting the Django Gunicorn server and the Flask reminder agent as separate, managed processes.

Expected Outcomes

- **Increased Medication Adherence:** A reduction in missed medication doses for users.
- **Improved Health Literacy:** Users will have access to personalized diet plans that are easy to follow.
- **Proactive Health Management:** The system will empower users to manage their health proactively through automated reminders and alerts.
- **A Scalable Platform:** A robust, microservices-based foundation for future feature development.

Future Scope

- **Implement Adaptive Reminders:** Enhance the reminder agent to use machine learning to analyze user behavior and suggest more optimal medication times.
- **Integrate Interactive Notifications:** Develop push notifications that allow users to mark a dose as "taken" or "snooze" directly.
- **Enable Grocery List Integration:** Connect the diet agent with grocery delivery services to allow users to automatically populate their shopping carts.
- **Expand to Weekly Diet Planning:** Upgrade the diet agent to generate comprehensive 7-day meal plans.

Conclusion

The Cura project represents a significant step towards leveraging AI for proactive and personalized healthcare. By automating the complex tasks of medication and diet management, Cura aims to provide a valuable tool that empowers users to lead healthier lives, ultimately contributing to the broader goal of well-being for all.